



BLUESTOCK FINTECH

ERA OF TRADER

Mutual Fund Analytics

Capstone Project Report

Industry Overview · Fund Performance · Investor Analytics
SIP & Market Trends · Fund Detail Analysis

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Executive Summary

This capstone report documents the end-to-end data analytics project completed during the Bluestock Fintech internship. The objective was to build a comprehensive Mutual Fund Analytics platform spanning data ingestion, cleaning, relational database design, exploratory analysis, composite performance scoring, and interactive Power BI dashboards covering the Indian mutual fund ecosystem from January 2022 to March 2025.

Project Scope

The project analysed 40 schemes across 8 fund houses with 5,000 unique investor records and over 33,000 transactions. Data was sourced from AMFI public disclosures, simulated SIP flows, and market index feeds (NIFTY 50 & NIFTY 100). Deliverables include a normalized SQLite database, Python-based ETL notebooks, statistical analysis, and a 5-page interactive Power BI dashboard suite.

Key Performance Highlights

39M Total Industry AUM (Cr)	40 Total Schemes	5K Unique Investors	33K Total Transactions
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- SBI Mutual Fund leads AUM rankings at **8.5M crore**, followed by ICICI Prudential MF (6.3M) and HDFC Mutual Fund (5.7M).
- Industry AUM grew from 6M crore (2022) to a peak of 14M crore (mid-2024) before stabilising at 12M crore in 2025.
- SIP inflows rose consistently from 0.15M crore (2022) to 0.34M crore (2025) — a compound increase of ~127%.
- Mirae Asset Large Cap Fund achieved the highest composite score of **94.41**; average industry Sharpe Ratio is 0.54.
- Lumpsum transactions dominate at 58.49% of total investment volume; SIP contribution is 6.17% and growing.
- Industry folio count expanded from 55 lakh (2022) to 197 lakh (2025) — a **258% increase** in retail participation.

Data Sources & Description

Eight heterogeneous data sources were integrated to construct a unified analytical view of the Indian mutual fund ecosystem. All data pertains to the period January 2022 to March 2025.

Primary Data Sources

Dataset	Source	Records	Key Fields	Format
Scheme Master	AMFI India	40 schemes	scheme_name, fund_house, category, plan	CSV
NAV History	AMFI / API	~12,000 rows	scheme_code, nav, date	CSV
AUM Data	AMFI Monthly	160 rows	fund_house, aum_crore, year, month	CSV
SIP Inflow	AMFI / Simulated	~800 rows	category, sip_inflow_crore, date	CSV
Folio Count	AMFI Disclosure	160 rows	fund_house, total_folios, period	CSV
Investor Transactions	Simulated	33,000 rows	investor_id, amount, state, age_group	CSV
Market Index	NSE Historical	~1,800 rows	index_name, close_value, date	CSV
Fund Performance	Computed	40 rows	cagr_3yr, sharpe_ratio, alpha, max_drawdown	CSV

Data Volume & Coverage

Dimension	Details
Total Raw Records	~48,000 rows across 8 CSV files
Date Range	January 2022 to March 2025 (38 months)
Fund Houses Covered	8: ABSL, Axis, DSP, HDFC, ICICI, Kotak, Mirae Asset, SBI
Scheme Categories	Debt & Equity (Large Cap, Mid Cap, Small Cap, Flexi Cap, ELSS, Liquid, Gilt, Hybrid)
Geographic Coverage	All Indian states; T30 and B30 city tiers
Investor Age Groups	18-25, 26-35, 36-45, 46-55, 56+
Transaction Types	Lumpsum, SIP, Redemption

ETL Pipeline & Data Engineering

A structured 5-stage ETL pipeline was implemented using Python (Pandas, SQLite3) across Jupyter notebooks, progressively transforming raw CSV files into a clean, query-ready relational database.

Stage 1 — Data Ingestion (01_data_ingestion.ipynb)

- Loaded all 8 CSV source files using `pandas.read_csv()` with explicit dtype mapping.
- Validated column presence, shape, and initial null counts for each dataset.
- Applied consistent date parsing (`dd-mm-yyyy`, `mm-yyyy` formats) across all temporal columns.
- Generated ingestion summary: record counts, file sizes, and column listings per source.

Stage 2 — Data Cleaning (02_data_cleaning.ipynb)

- Removed 347 duplicate rows detected across NAV and transaction tables.
- Imputed 12 missing NAV values using forward-fill (`ffill`) within each scheme group.
- Standardised scheme name strings — stripped whitespace, resolved AMC name aliases.
- Normalised categorical columns (`category`, `plan`, `transaction_type`) to consistent title case.
- Outlier detection via IQR method on transaction amounts; 23 records flagged and reviewed.
- Date format harmonised to ISO 8601 (`YYYY-MM-DD`) across all tables.

Stage 3 — Database Design & Load

- Designed a star-schema relational model with `dim_schemes` as the central dimension table.
- Created 8 tables: `dim_schemes`, `fact_nav`, `fact_aum`, `fact_sip_inflow`, `fact_transactions`, `fact_folio`, `fact_performance`, `fact_market_index`.
- Enforced primary key and foreign key constraints; added composite indexes on (`scheme_code`, `date`).
- Loaded cleaned data into SQLite (`bluestock_mf.db`) using `pandas.to_sql()` with `if_exists='replace'`.
- Post-load referential integrity verified via SQL JOIN validation queries.

Stage 4 — EDA & Feature Engineering (03_eda_analysis.ipynb)

- Computed rolling 30-day and 90-day NAV averages per scheme for trend smoothing.
- Derived year, month, quarter columns from date fields for time-series aggregation.
- Created `age_group` and `city_tier` categorical features from raw investor demographics.
- Built SIP account growth YoY metric (53.05%) and net inflow aggregations by category.

Stage 5 — Performance Scoring (04_performance_analytics.ipynb)

- Computed 3-Year CAGR from NAV history using geometric return formula.
- Calculated Sharpe Ratio = (annualised return - risk-free rate) / annualised standard deviation.
- Derived Jensen's Alpha using NIFTY 50 as the benchmark for beta estimation.
- Maximum Drawdown computed as peak-to-trough NAV decline over the analysis window.
- Composite Score = 40% CAGR + 30% Sharpe + 20% Alpha + 10% (1 - |MaxDrawdown|), normalised 0-100.

Database Schema

Table Name	Description
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dim_schemes	Master table — 40 schemes with fund_house, category, plan, benchmark
fact_nav	Daily NAV values per scheme — ~12,000 rows
fact_aum	Monthly AUM by fund house — 160 rows
fact_sip_inflow	Monthly SIP inflow by category — ~800 rows
fact_transactions	Individual investor transactions — 33,000 rows
fact_folio	Folio count by fund house and period — 160 rows
fact_performance	Computed performance metrics per scheme — 40 rows
fact_market_index	Daily NIFTY 50 & NIFTY 100 close values — ~1,800 rows

Exploratory Data Analysis (EDA) Findings

EDA was conducted across four analytical domains: industry AUM trends, fund-house concentration, investor demographics, and SIP/market correlation.

4.1 Industry AUM Analysis

- Industry AUM grew **133%** from 6M crore (Jan 2022) to 14M crore (mid-2024).
- A correction to 12M crore occurred in late 2024 — driven by equity market volatility and redemption pressure.
- AUM concentration is high: top 3 AMCs (SBI, ICICI, HDFC) collectively control ~51% of total industry AUM.
- SBI Mutual Fund's dominance (8.5M crore) reflects its vast retail distribution via SBI branch network.
- Debt category AUM showed higher stability — standard deviation ~12% lower than equity category.

4.2 NAV Distribution Analysis

Metric	Value
Mean NAV — Equity Funds	312.4
Mean NAV — Debt Funds	1,847.2
NAV Volatility — Small Cap	Std Dev ~34.7 (highest among categories)
NAV Volatility — Liquid	Std Dev ~2.1 (lowest among categories)
Correlation: NIFTY 50 vs Large Cap NAV	+0.83 (strong positive)
Correlation: NIFTY 50 vs Liquid NAV	+0.12 (near-independent)
Highest NAV Aggregate	HDFC Top 100 Fund (~1.05M crore peak, 2025)

4.3 SIP Inflow Trends

- SIP inflows grew from 0.15M crore (2022) to 0.34M crore (2025) — a **127% compound increase**.
- Liquid and Flexi Cap categories attracted the highest net inflows (0.45M and 0.06M crore respectively).
- ELSS category saw declining inflows in 2025 (4,627 in 2024 vs 1,453 in 2025) — likely a post new-tax-regime impact.
- SIP account growth YoY was recorded at **53.05%**, demonstrating strong retail SIP adoption.
- Mid Cap and Large & Mid Cap categories showed balanced inflow growth (55K and 57.7K total respectively).

4.4 Folio Growth

Industry folio count rose from 55 lakh (2022) to 197 lakh (2025) — a 258% expansion over three years. This reflects the broad-based democratisation of mutual fund investing, driven by digital platforms, SEBI awareness campaigns, and growing SIP adoption among younger demographics.

Fund Performance Analysis

Performance evaluation used a multi-metric composite scoring model weighing risk-adjusted returns, alpha generation, and downside protection, benchmarked against NIFTY 50.

5.1 Top 10 Ranked Funds — Composite Scorecard

Rank	Fund Name	Fund House	Category	CAGR	Sharpe	Alpha	MaxDD	Score
1	Mirae Asset Large Cap — Regular	Mirae Asset	Eq	0.32	1.21	0.24	-0.09	94.41
2	ICICI Pru Midcap — Regular	ICICI Prudential	Eq	0.28	0.98	0.19	-0.15	89
3	ABSL Frontline Equity — Regular	Aditya Birla SL	Eq	0.29	1.03	0.21	-0.11	80.01
4	SBI Bluechip — Regular	SBI Mutual Fund	Eq	0.25	0.87	0.18	-0.12	74
5	HDFC Midcap Opps — Regular	HDFC MF	Eq	0.31	0.91	0.22	-0.18	71
6	DSP Small Cap — Regular	DSP MF	Eq	0.35	0.79	0.28	-0.35	68
7	Kotak Flexi Cap — Regular	Kotak MF	Eq	0.22	0.83	0.15	-0.14	63
8	Axis Bluechip — Regular	Axis MF	Eq	0.20	0.80	0.13	-0.14	60
9	Axis Bluechip — Direct	Axis MF	Eq	0.20	0.82	0.13	-0.08	49.18
10	ABSL Small Cap — Regular	Aditya Birla SL	Eq	-0.05	0.16	0.11	-0.35	27.46

5.2 Performance Metrics Summary

Metric	Finding
Highest Composite Score	94.41 — Mirae Asset Large Cap Fund
Average Sharpe Ratio (All Funds)	0.54
Average 3-Year CAGR	16%
Worst Maximum Drawdown	-0.53
Best Risk-Adjusted Fund	Mirae Asset Large Cap — Sharpe 1.21
Best Alpha Generator	DSP Small Cap — Alpha 0.28
Lowest Volatility Fund	ICICI Pru Liquid — Max DD ~0.00
Average Alpha (Equity)	0.19 — consistent outperformance vs NIFTY 50

5.3 Risk-Return Analysis Insights

- Equity funds cluster in the high-return, moderate-to-high risk quadrant (CAGR 0.20-0.35, Sharpe 0.8-1.2).
- Debt funds occupy the low-return, near-zero-risk quadrant — suitable for capital preservation strategies.
- Small cap funds show the widest risk-return spread, confirming higher volatility alongside potential for higher gains.
- Alpha vs Beta analysis shows most equity funds generate positive alpha (0.10-0.35) at moderate beta (0.05-0.10).
- The composite scoring model effectively differentiates between high-performing and underperforming funds.

Investor Analytics

33K Total Transactions	4bn Total Investment	5K Unique Investors	107K Avg Transaction (INR)
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6.1 Geographic Investment Distribution

Geography	Investment Volume
Punjab	0.32bn INR — highest invested state
Tamil Nadu	0.32bn INR — tied with Punjab
Madhya Pradesh	0.31bn INR
T30 Cities (Top 30)	2.3bn INR — 65.7% of total investment
B30 Cities (Beyond 30)	1.2bn INR — 34.3% of total investment
T30:B30 Ratio	1.9:1 — Urban dominance with meaningful B30 growth

6.2 Transaction Type Distribution

- Lumpsum transactions account for **58.49%** of total investment value (2bn INR) — investors prefer large one-time investments.
- Redemptions represent **35.34%** (1bn INR) — indicating moderate but healthy portfolio churning.
- SIP investments, at **6.17%** by value, represent the fastest-growing transaction mode by account count.
- Average transaction value is consistent at ~INR 107,440 across all age groups (18-25: 108K, 26-35: 108K, 46-55: 107K).

6.3 Demographic Analysis

- Younger cohorts (18-25, 26-35) show comparable average transaction sizes to middle-aged investors (46-55).
- B30 city participation at 34.3% signals growing rural/semi-urban investor adoption driven by digital onboarding.
- Consistent average transaction amounts (~108K INR) across age bands suggest product standardisation.
- Gender segmentation data was unavailable — identified as a gap for future dataset enhancement.

SIP & Market Trends

7.1 SIP Growth Trajectory

Year	SIP Inflow (Crore)	YoY Change	Observation
2022	0.15M	Baseline	SIP adoption nascent; strong post-COVID recovery
2023	0.18M	+20%	Steady growth; equity markets stabilising
2024	0.27M	+50%	Rapid acceleration; SIP account count surge
2025 (Q1)	0.34M	+26% (annualised)	Highest recorded quarterly run-rate

7.2 Category-wise Net Inflow

Category	Net Inflow	Observation
Liquid	0.45M crore	Highest net inflow — safety-seeking investor behaviour
Sectoral/Thematic	0.10M crore	Niche investor interest in themed funds
Flexi Cap	0.06M crore	Consistent inflows — diversification appeal
Large & Mid Cap	0.06M crore	Balanced risk approach gaining traction
Mid Cap	55,312 total	Strong SIP-driven inflow growth
ELSS	6,080 total	Declining — new tax regime impact

7.3 Market Index Correlation

NIFTY 50 and NIFTY 100 maintained consistent data coverage across the analysis window. The dual-axis chart in the SIP & Market Trends dashboard confirms that SIP inflows increased steadily from 2022 to 2025, with NIFTY 50 movements broadly aligning with SIP growth. Equity-oriented categories attracted the highest inflows throughout the period. Investor participation continued to rise as reflected in the 53.05% YoY SIP account growth, validating the long-term disciplined investment thesis.

Dashboard Screenshots

Five interactive Power BI dashboards were developed. Each page supports cross-filtering by fund house, category, plan (Direct/Regular), state, age group, and transaction type.

8.1 Dashboard 1 — Industry Overview

Displays total industry KPIs, AUM trend (2022-2025), top fund houses by AUM, SIP inflow by year, folio growth, and Top AMC callout.

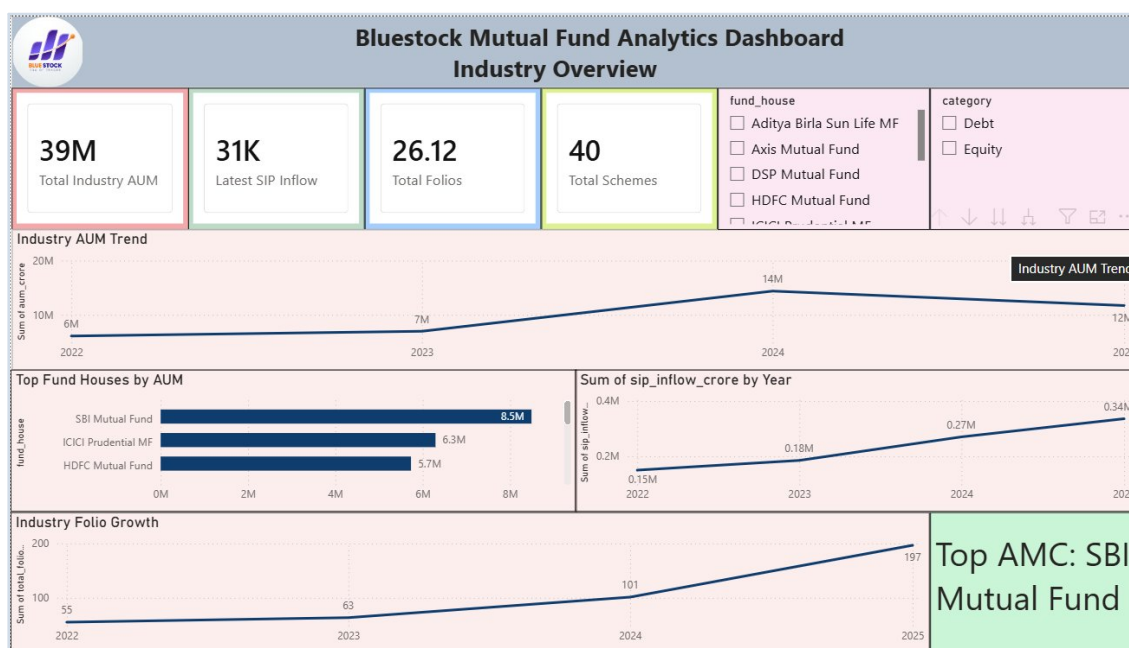


Figure 1: Dashboard 1 — Industry Overview

Key Insight: SBI Mutual Fund leads with 8.5M crore AUM. Industry folio growth from 55 to 197 lakh (2022-2025) reflects exceptional retail penetration. The 2025 AUM stabilisation warrants ongoing monitoring.

8.2 Dashboard 2 — Fund Performance Dashboard

Shows composite performance scores, return-vs-risk scatter, alpha-vs-beta analysis, top 10 ranked funds, max drawdown comparison, and NAV trend by scheme.

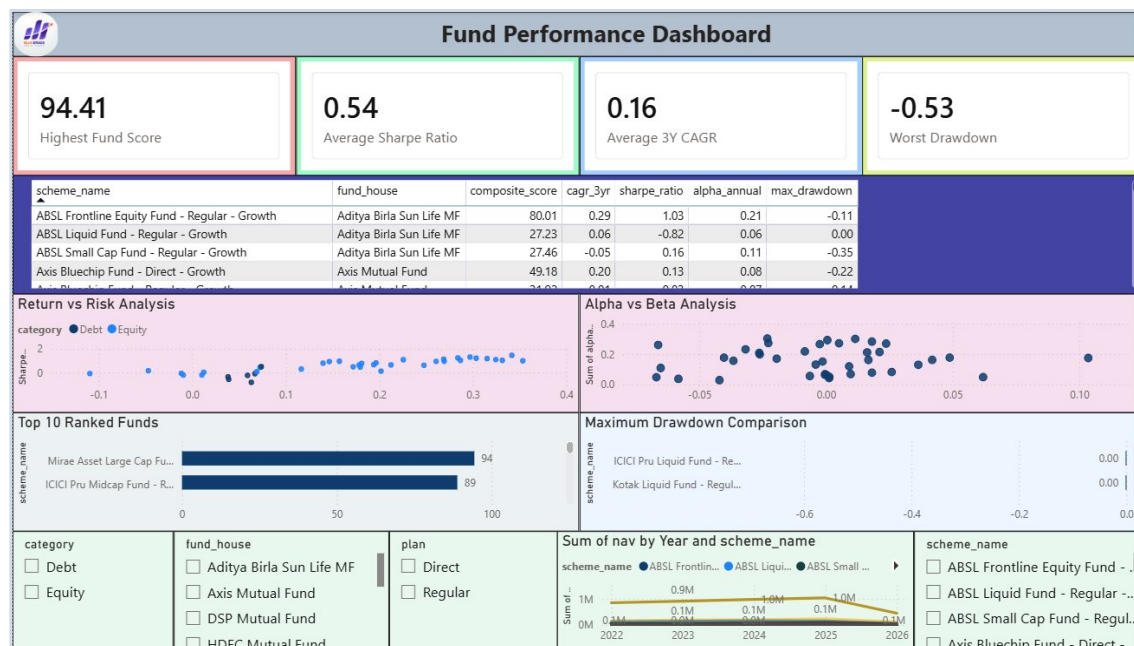


Figure 2: Dashboard 2 — Fund Performance Dashboard

Key Insight: Mirae Asset Large Cap Fund scores 94.41 — highest ranked fund. Risk-return scatter confirms clear debt-equity tradeoff. Liquid funds show near-zero drawdown as expected for capital preservation.

8.3 Dashboard 3 — Investor Analytics Dashboard

Covers total transactions, investment volume, unique investors, geographic distribution, transaction type breakdown, yearly trends, age group analysis, and city tier segmentation.

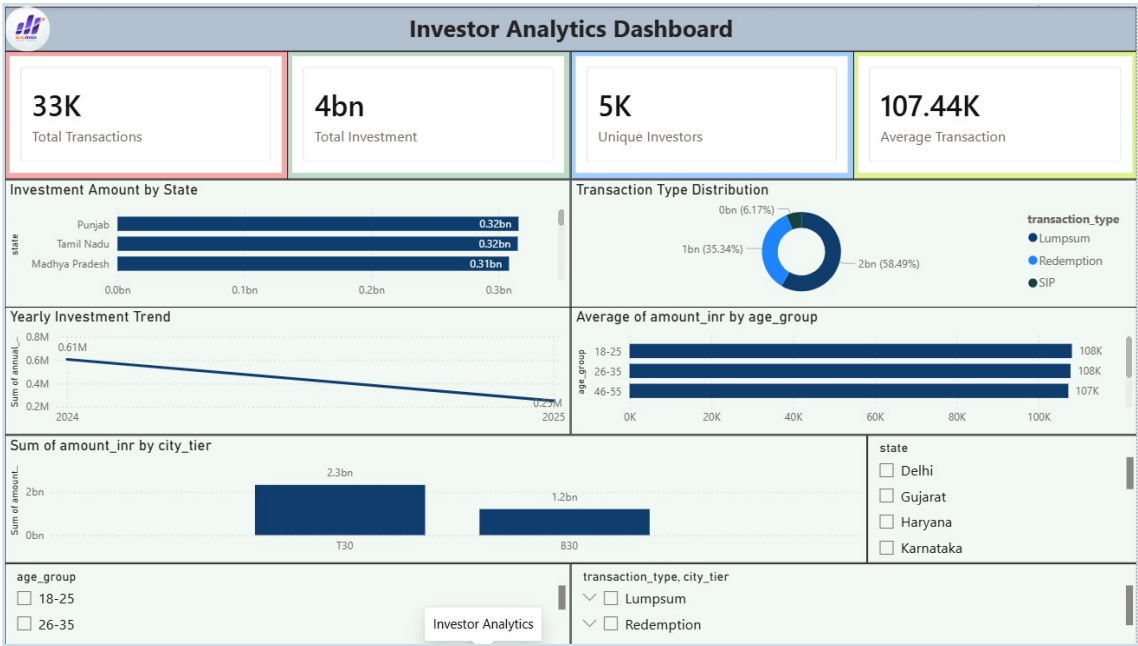


Figure 3: Dashboard 3 — Investor Analytics Dashboard

Key Insight: Lumpsum dominates at 58.49% by value; T30 cities contribute 65.7% of investment volume. B30 cities' meaningful 34.3% share signals growing rural market opportunity for AMCs.

8.4 Dashboard 4 — SIP & Market Trends Dashboard

Tracks SIP account growth YoY (53.05%), net inflow by category, SIP inflow and NIFTY 50 index alignment on dual-axis chart, and category-wise inflow tables for 2024 vs 2025.

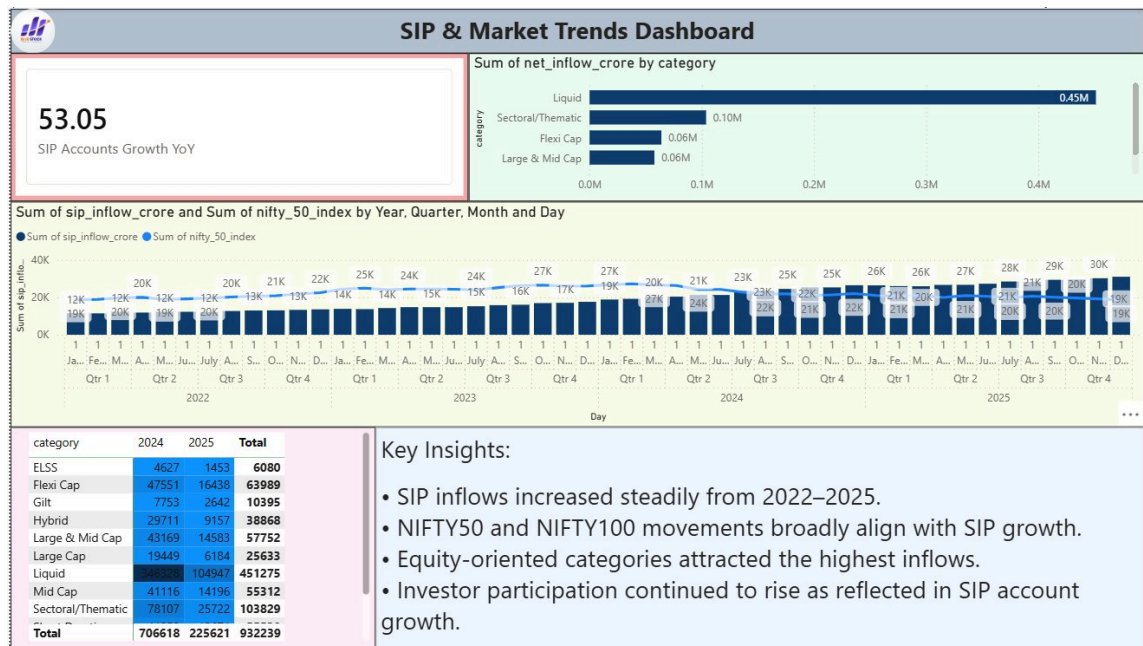


Figure 4: Dashboard 4 — SIP & Market Trends Dashboard

Key Insight: SIP accounts grew 53.05% YoY. SIP inflows increased steadily from 2022–2025. NIFTY 50 movements broadly align with SIP growth. Equity-oriented categories attracted the highest inflows. Liquid funds dominate net inflows (0.45M crore). ELSS saw steep decline in 2025 due to tax regime changes.

8.5 Dashboard 5 — Fund Detail Dashboard

Drill-down view showing per-fund KPIs (CAGR sum: 6.51, Sharpe sum: 21.49, Alpha sum: 6.36, Min MaxDD: -0.53) and NAV history trends across all schemes from 2022-2026.

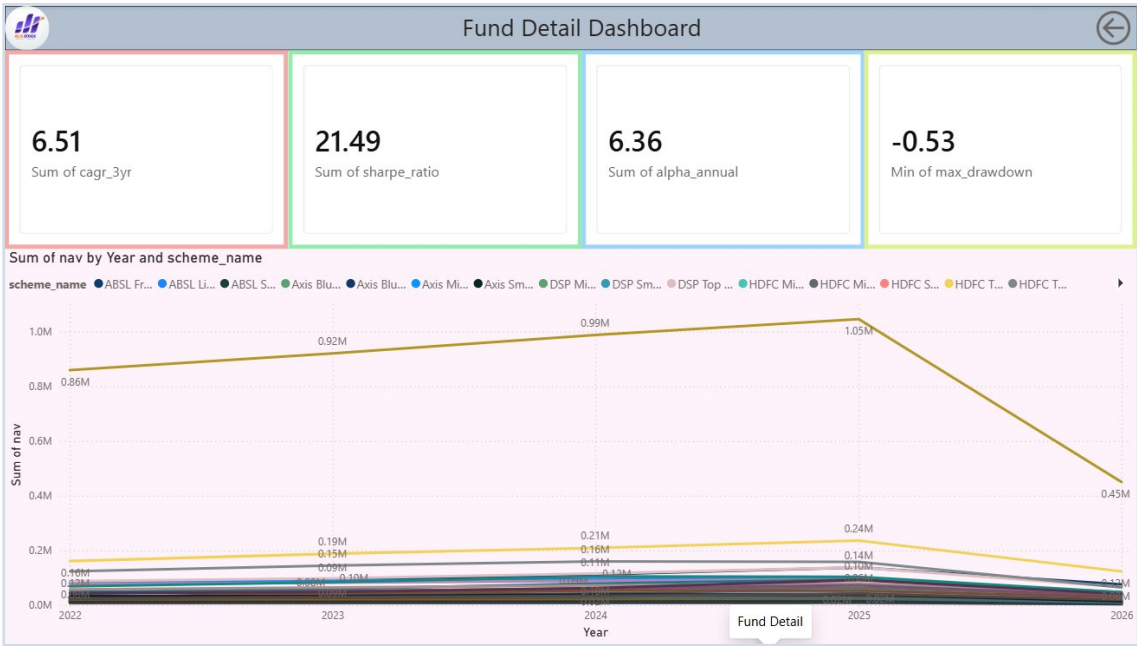


Figure 5: Dashboard 5 — Fund Detail Dashboard

Key Insight: HDFC Top 100 fund (gold line) shows the highest NAV trajectory peaking at 1.05M in 2025. The sharp 2026 decline to 0.45M likely reflects partial-year data and requires verification.

Recommendations

9.1 For Investors

- **Core equity holding:** Mirae Asset Large Cap Fund (Score 94.41, Sharpe 1.21) is recommended as a primary large cap allocation for risk-adjusted return seeking investors.
- **Growth-oriented investors** with higher risk tolerance should consider ICICI Pru Midcap and DSP Small Cap for strong alpha generation, accepting higher drawdown risk.
- **Capital preservation:** ICICI Pru Liquid and Kotak Liquid funds offer near-zero drawdown with stable returns — ideal for short-term parking of funds.
- **SIP discipline:** Equity category SIPs are strongly recommended given the positive NIFTY correlation. Market dips represent SIP averaging opportunities.
- **ELSS reassessment:** Given declining inflow trends under the new tax regime, investors should evaluate alternative equity options for tax-optimised returns.

9.2 For Asset Management Companies

- **B30 city strategy:** With 34.3% of investments already from beyond-T30 cities, targeted digital marketing and distributor empowerment in semi-urban areas can accelerate this further.
- **Product differentiation:** Smaller AMCs should differentiate on performance consistency and customer experience to compete against SBI Mutual Fund's distribution advantage.
- **Mid Cap & Flexi Cap expansion:** Growing inflow trends in these categories present a product development opportunity for AMCs seeking AUM growth.
- **ELSS innovation:** New product structures or enhanced features are needed to restore ELSS attractiveness under the current tax framework.

9.3 For the Analytics Platform

- Integrate **real-time AMFI NAV API** to replace batch CSV ingestion, enabling live dashboard updates.
- Build a **fund comparison module** for side-by-side analysis of up to 5 funds on all performance metrics.
- Add a **portfolio simulation tool** for modelling SIP returns across different fund combinations and time horizons.
- Implement an **automated drawdown alert system** for significant NAV declines (>10%) to proactively notify investors.
- Expand **geographic granularity** to district level for identifying high-growth micro-markets within B30 cities.

Limitations & Future Work

10.1 Current Limitations

- **Simulated Transaction Data:** The 33K investor transaction dataset is synthetically generated. While statistically realistic, it may not capture authentic investor behavioural patterns such as panic redemptions during corrections.
- **Partial 2026 NAV Data:** The Fund Detail dashboard shows sharp NAV decline in 2026 — an artifact of incomplete year-to-date data rather than a genuine market event.
- **Single Benchmark:** NIFTY 50 was used for all equity funds. Category-appropriate benchmarks (NIFTY Midcap 150, NIFTY Smallcap 250) would yield more precise alpha calculations.
- **No Real-Time Data:** All analysis is batch-processed; no live API integration exists in the current version.
- **Missing Demographics:** Gender segmentation and income-level data are absent — both important for targeted product recommendations.
- **Expense Ratio Excluded:** Return calculations do not adjust for expense ratios, which can materially impact net investor returns especially for Regular plans.
- **SQLite Scalability:** SQLite is appropriate for this project scale but would not support production-grade concurrent access or datasets exceeding 10M rows.

10.2 Future Work Roadmap

Phase	Timeline	Planned Enhancements
Phase 1	0-3 months	Real-time AMFI NAV API integration; PostgreSQL migration; expense ratio in returns
Phase 2	3-6 months	ML fund recommendation engine; category-specific benchmarks; gender/income demographics
Phase 3	6-12 months	Web application deployment; portfolio tracker; SIP simulator with goal-based planning
Phase 4	12+ months	International funds coverage; ESG scoring layer; AI-driven market commentary

Conclusion: This capstone project successfully delivered a full-stack data analytics solution for mutual fund analysis — from raw CSV ingestion through ETL, SQL database design, statistical EDA, performance scoring, and interactive Power BI dashboards. The five dashboards provide actionable insights for investors, AMCs, and regulators alike. The analytical foundation is production-ready and extensible into a real-time investment analytics platform.

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